MITRE ATT&CK Navigator Analysis Report

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1. Introduction

The MITRE ATT&CK framework is a globally recognized knowledge base that categorizes adversary behavior based on real-world observations. It provides a structured matrix of tactics and techniques used by threat actors, enabling cybersecurity professionals to analyze, detect, and mitigate threats effectively.

To facilitate exploration and analysis of this framework, MITRE provides the ATT&CK Navigator, a web-based tool that allows users to visualize techniques, compare threat groups, and create customized analytical layers.

This report demonstrates practical usage of the ATT&CK Navigator through multiple use cases, highlighting my ability to interpret and apply ATT&CK data for threat intelligence analysis.

2. Objectives

- Develop familiarity with the ATT&CK Navigator interface and features
- Apply ATT&CK techniques to real-world threat groups
- Perform comparative analysis using Navigator layers
- Demonstrate hands-on capability in cybersecurity threat modeling

3. Tools and Resources

- ATT&CK Navigator (Web Version) : https://mitre-attack.github.io/attack-navigator/
- GitHub Repository: https://github.com/mitre-attack/attack-navigator
- Supporting tutorials and documentation
    1. Introduction to ATT&CK Navigator - YouTube
    2. How to use the MITRE ATT&CK Navigator - YouTube

4. Methodology

The analysis was conducted using the web-based ATT&CK Navigator. The following key functionalities were explored:

1. Creating new layers
2. Opening existing layers
3. Combining layers for comparative analysis

Two threat groups were selected for comparison:

- APT18
- APT38

These groups were chosen due to their distinct operational objectives, allowing meaningful comparison.

5. Use Case Implementation

5.1 Creating New Layers

New layers were created to represent individual threat groups.

- A layer was created for **APT18**, assigning:
    - Score: 1
    - Color: Red

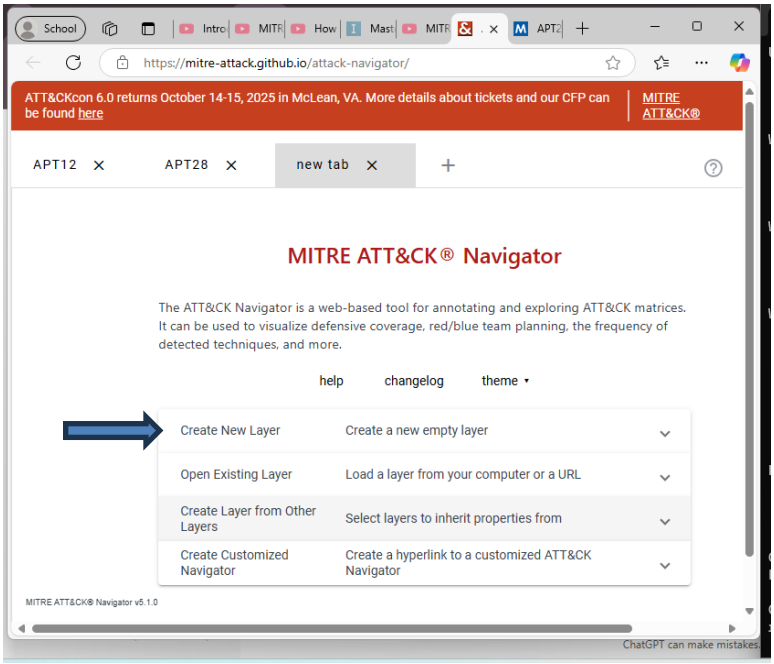

Figure 1: Creating a new layer for APT18

Named the layer name APT 18 .Under threat group scroll down and selected “search & multi select” option. After that clicked select button beside APT 18 .

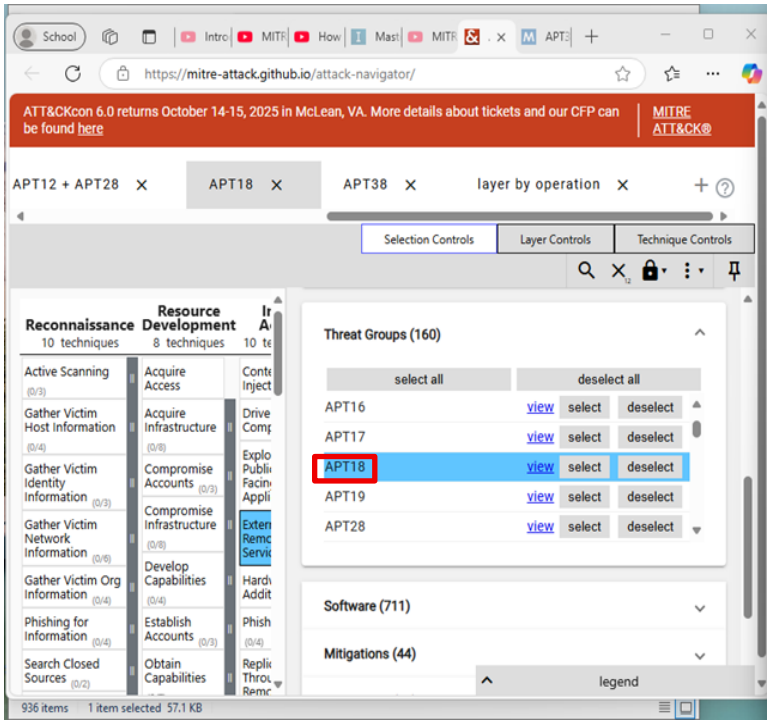

Figure 2: Select APT18

Set score 1 and color red for APT18 techniques layer as shown in Figure 3 and 4.

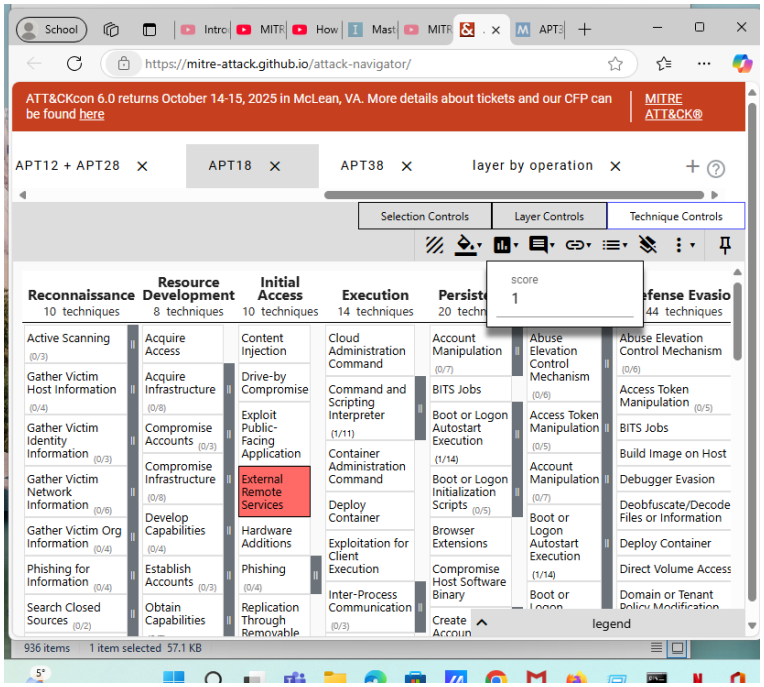

Figure 3: Set score for APT18

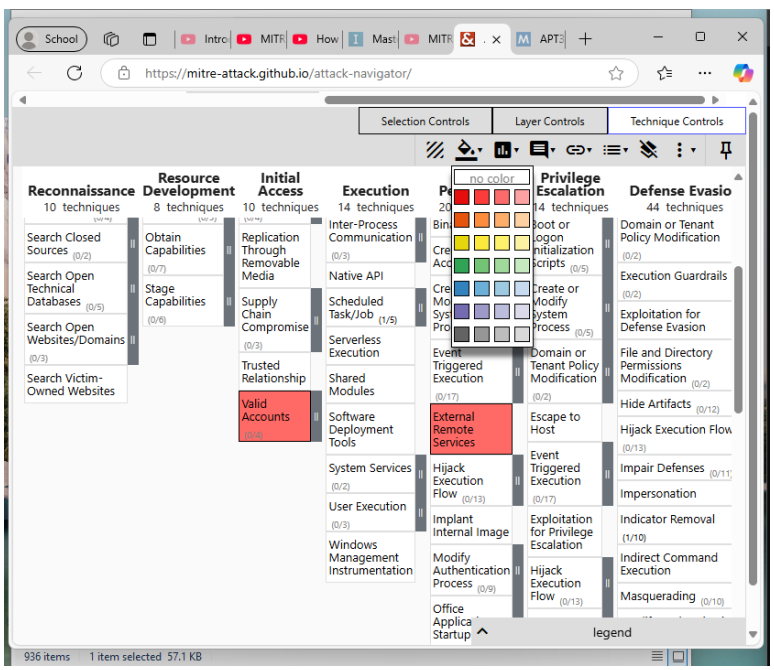

Figure 4:Set color for APT18

- A second layer was created for **APT38**, assigning:
  - Score: 2
  - Color: Yellow

Repeat the process to create APT 38 layer. Figures 5 and 6 illustrate that, for the APT38 layer, a score of 2 has been assigned and is represented using the color yellow.

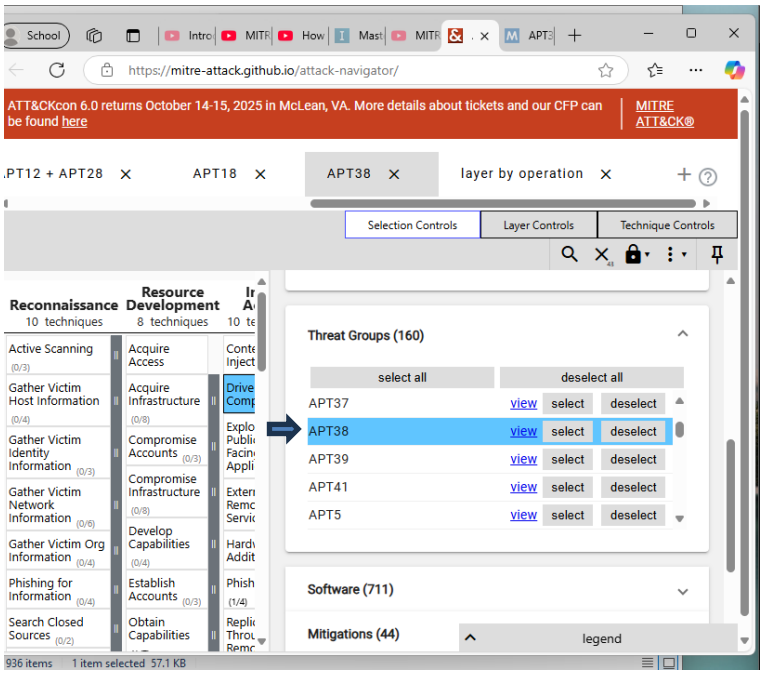

Figure 5: Select APT38 Technique

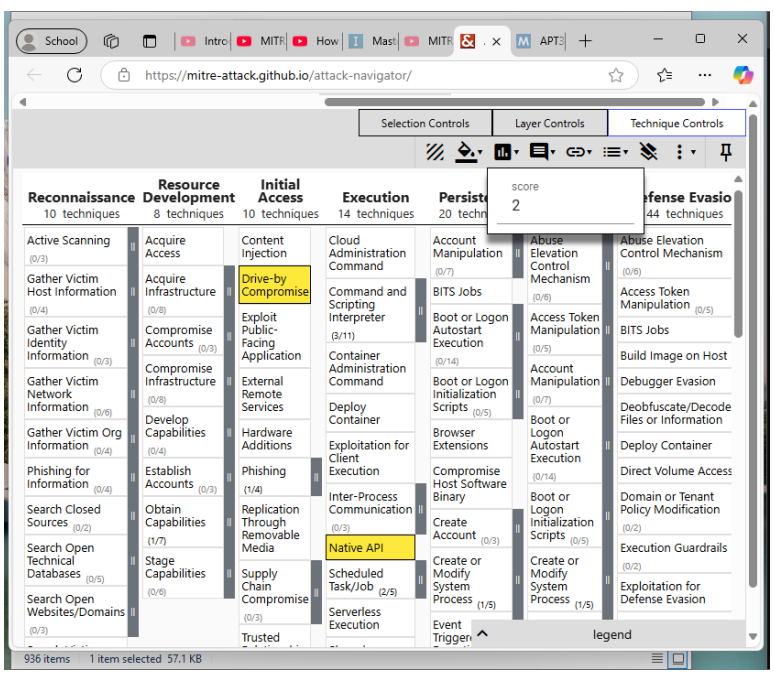

Figure 6: Set score & color for APT38

Analysis:
This step demonstrates how ATT&CK techniques can be mapped to specific adversary groups. Assigning scores and colors enables visual prioritization and differentiation of techniques across threat actors.

5.2 Opening Existing Layers

Previously created layers (APT12 and APT28) were exported as JSON files and re-imported into the Navigator.

Process:

- Export layers using *Layer Controls → Export*
- Download as JSON
- Re-import using *Open Existing Layer → Upload from Local*

For this task I created APT 12 and APT 28 layer. After creating these two layers I downloaded both file as Json in my computer. To download I clicked “Layer Controls” . There I chose “Export” and then clicked “download all layers as Json ” as shown in Figure 7.

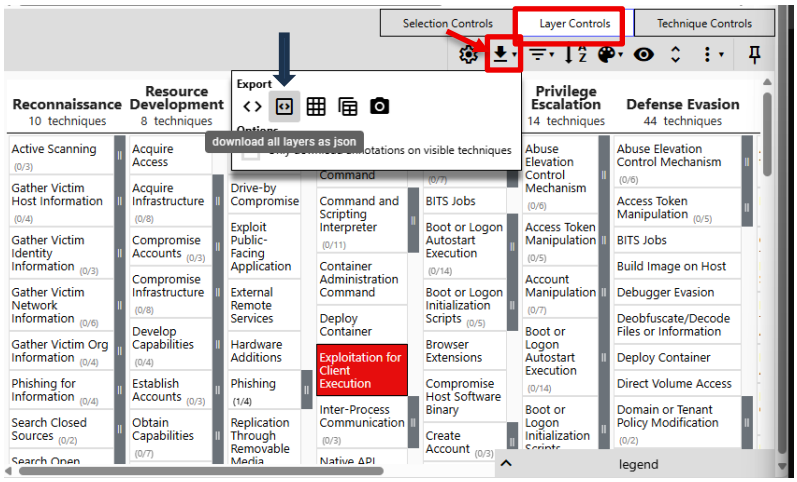

Figure 7: Downloading layers as JSON file

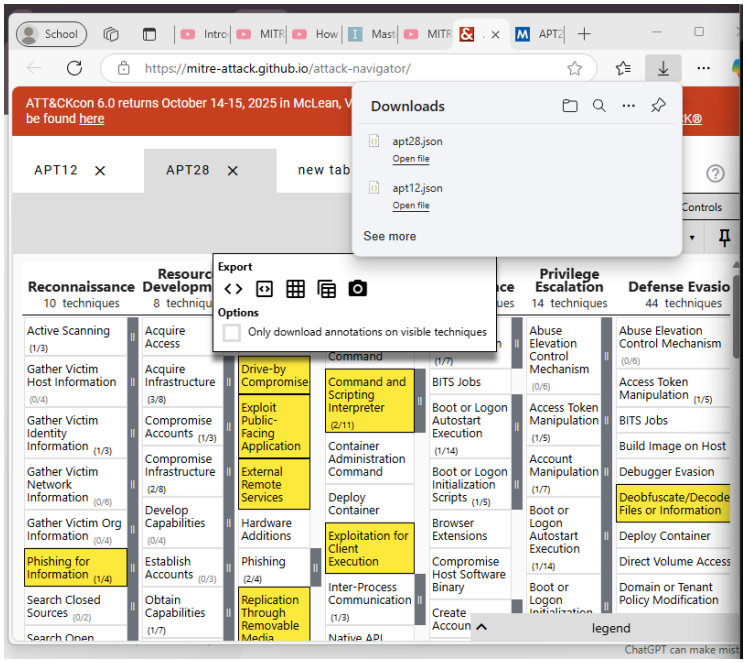

Figure 8: Layers downloaded successfully in my computer

Figure 8 depicts, apt12 and apt 28 Json files were downloaded to my computer.

Selected the option “Open Existing Layer” and load them in MITRE ATT&CK navigator .

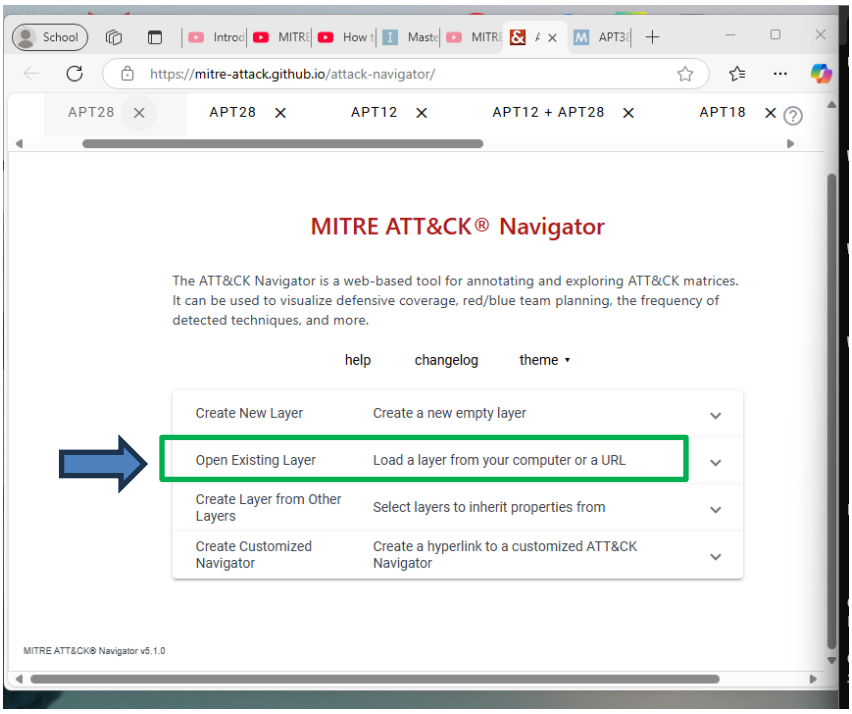

Figure 9:Open Existing Layer

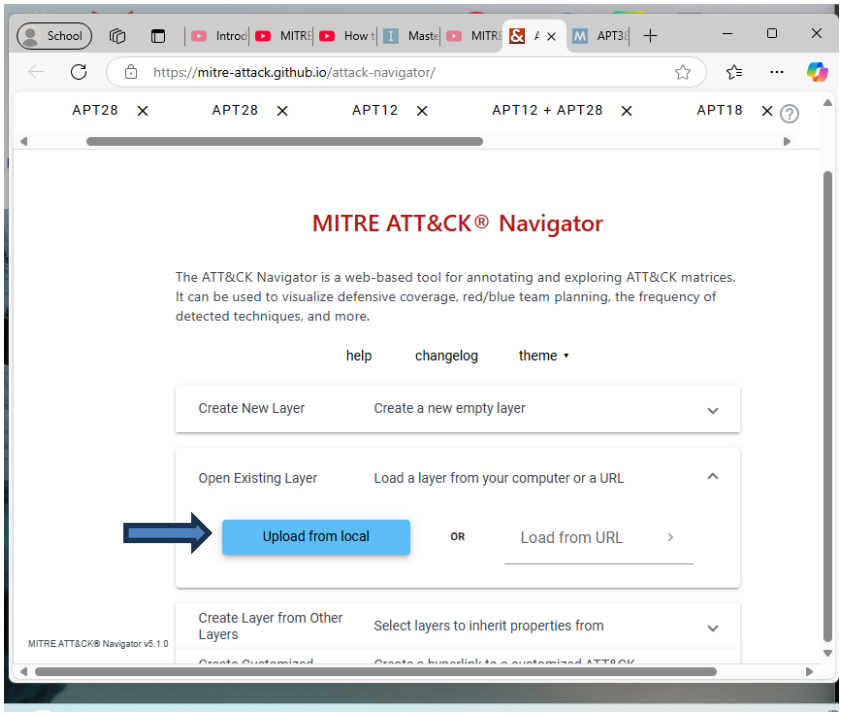

Figure 10 :File upload

Select Upload from local to upload the Json files.

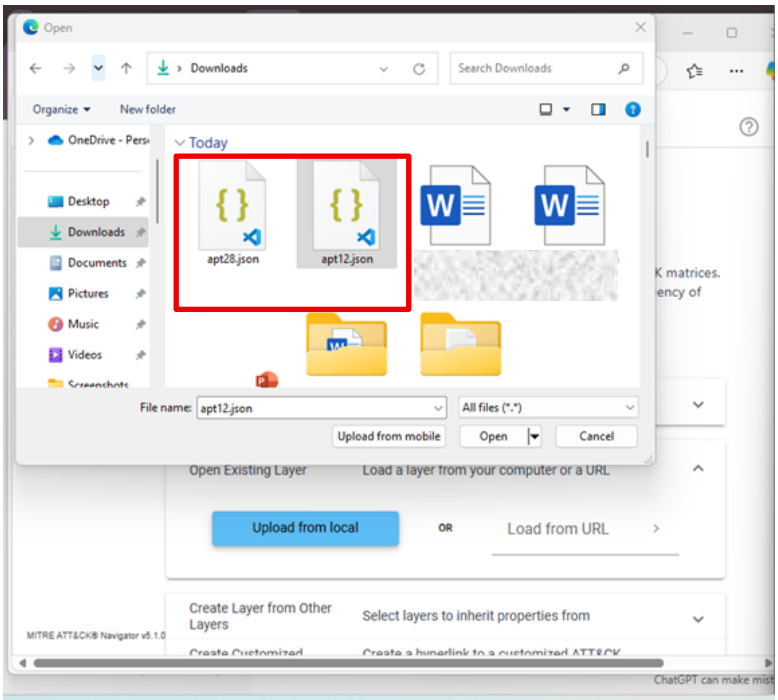

Figure 11:APT12 JSON File selected to upload

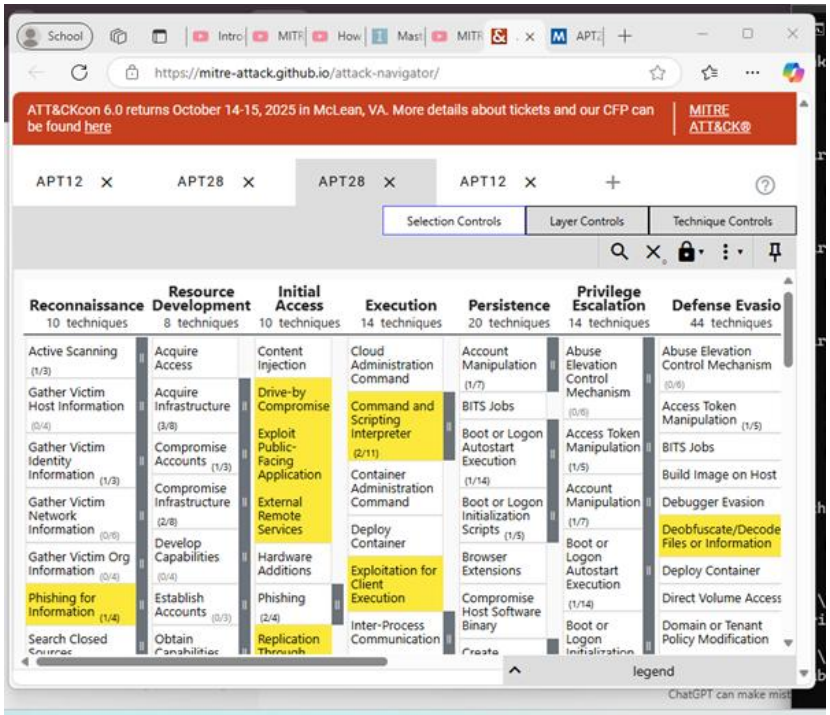

Figure 12:Uploaded the files successfully

APT28 & APT12 JSON files were uploaded.

Analysis:
This functionality supports collaboration and reproducibility in cybersecurity analysis. Analysts can share layer configurations across teams, improving consistency in threat intelligence workflows.

5.3 Creating Layers from Other Layers (Comparative Analysis)

A combined analytical layer was created to compare:

- APT18 (f)
- APT38 (g)

Important Observation:
This feature was not available in ATT&CK version 4, requiring a switch to **version 16**.

Configuration:

- APT18 → Score 1 (Red)
- APT38 → Score 2 (Yellow)

- Combined Layer (APT18 + APT38) → Score 3 (Green)

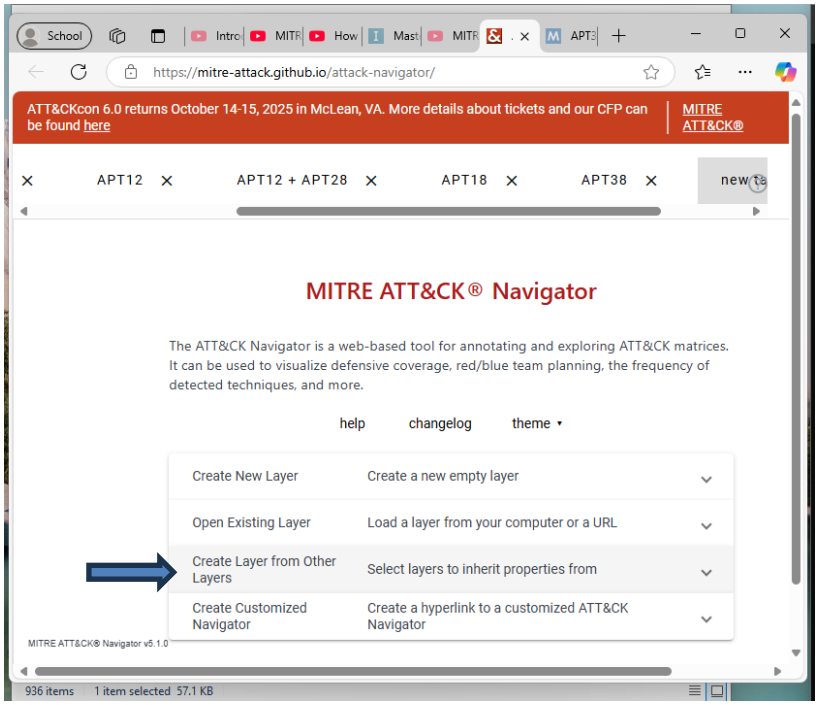

Figure 93:Create layer from Other Layers

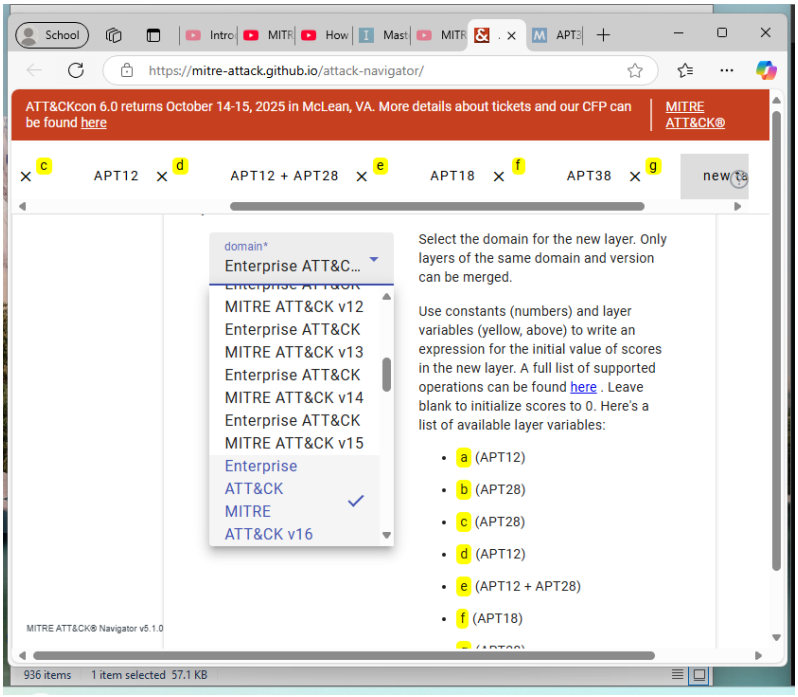

Figure 14:MITRE ATT&CK version 16

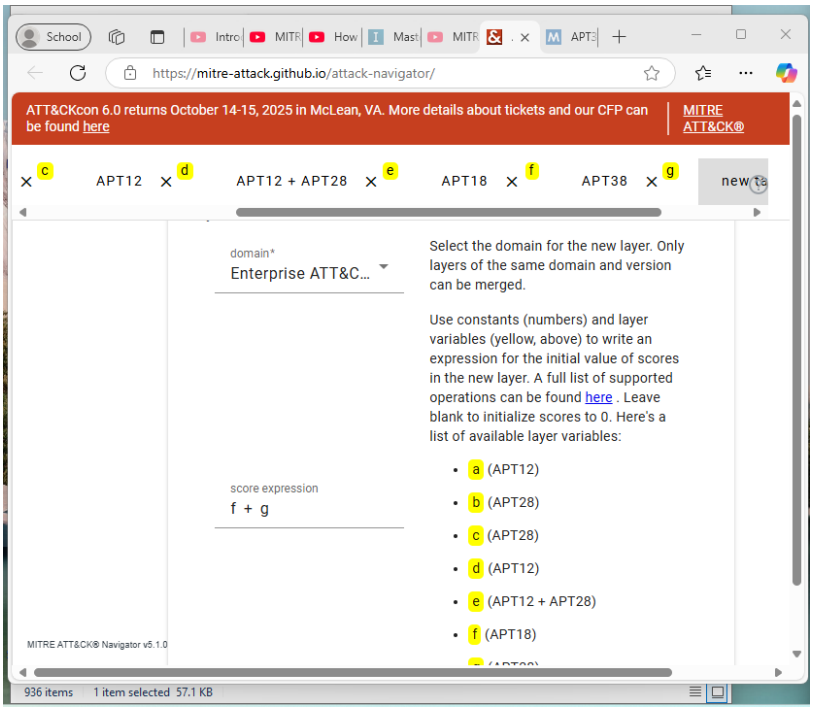

Figure 15:Setting score expression for f and g layers

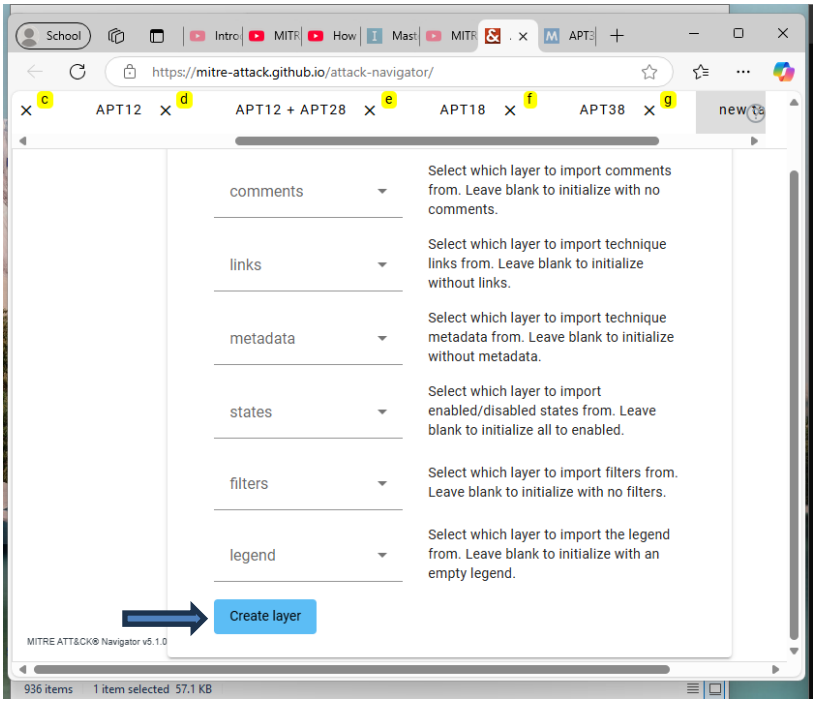

Figure 16:Create Layer

Clicked “Create Layer” .

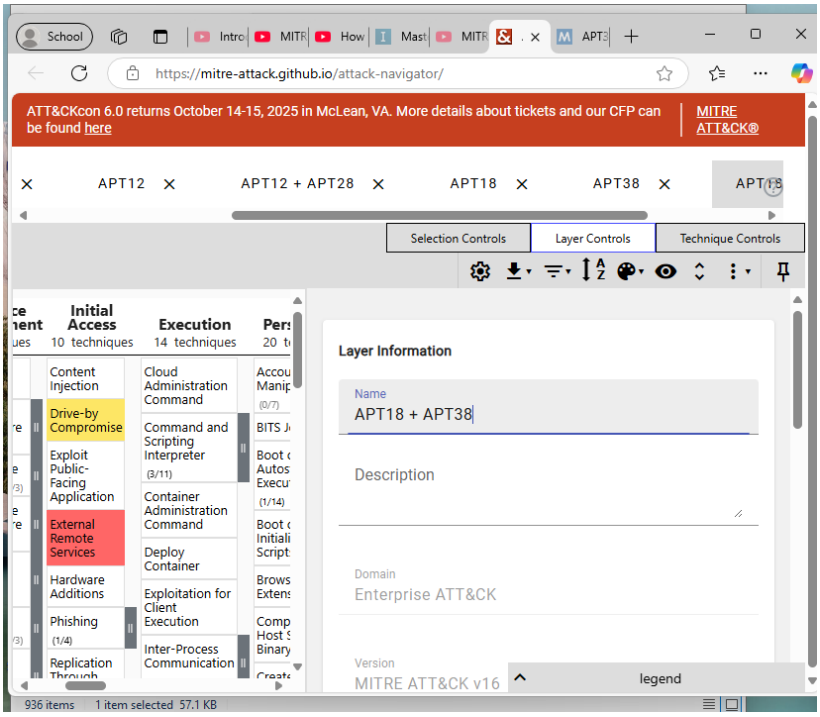

Figure 17:Giving name "APT18+38" for this layer

I gave name "APT18+38" for this Layer.

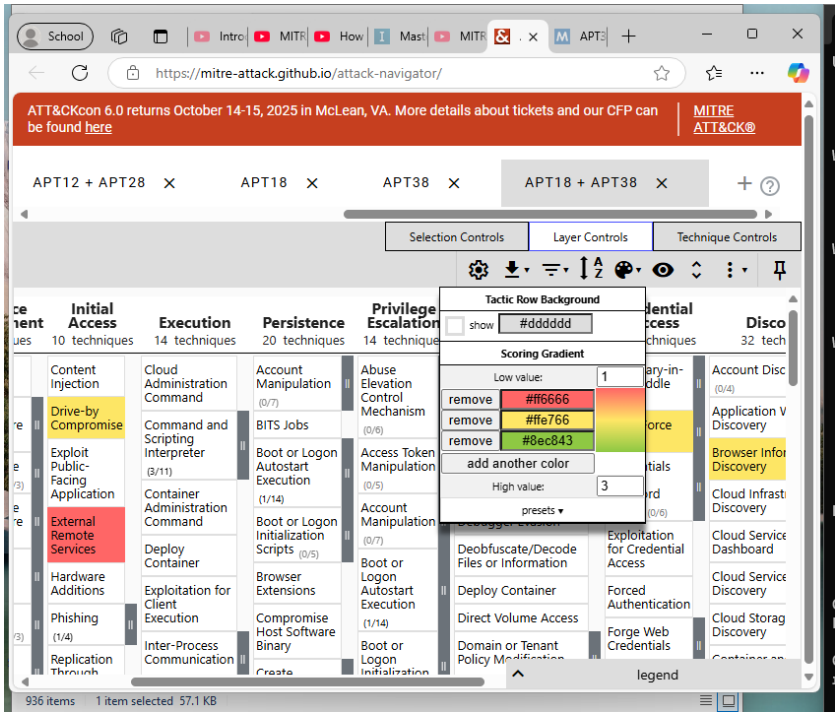

Figure 18:Setting score and color

In Figure 18 - 19, I set Low value “1” and High value "3” . Selected these three different colors to identify and differentiate these layers.

APT 18(f) : Score “1” and color Red.

APT 38 (g):Score “2” and color Yellow

APT 18 + APT 38 (f + g) : Score “3” and colour Green.

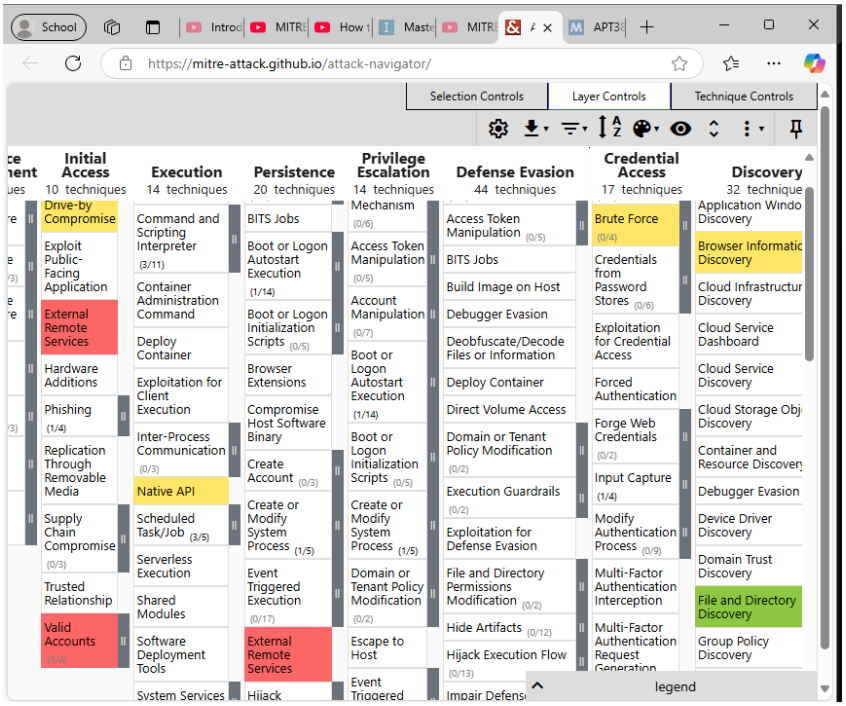

Figure 19:Successfully created layer from other layers

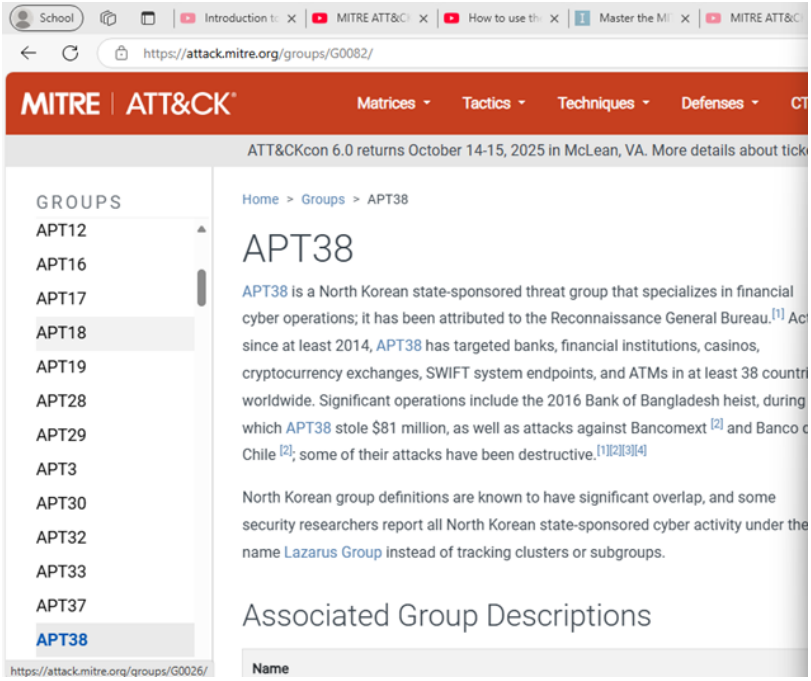

Figure 20:Techniques for threat groups

Analysis:
This step highlights the analytical strength of ATT&CK Navigator:

- Overlapping techniques reveal shared adversary behaviors
- Unique techniques highlight differences in operational goals
- Scoring allows prioritization of combined threat coverage

This comparative approach is particularly useful for:

- Threat hunting
- Detection engineering
- Security control gap analysis

6. Validation of Techniques

To ensure the accuracy and completeness of the mapped techniques, those associated with APT38(Fig 20) were systematically validated against official MITRE ATT&CK data sources. This process involved cross-verifying each technique included in the Navigator layer with authoritative references to confirm correct mapping and coverage.

Analysis:
This validation step strengthens the reliability of the analysis by ensuring data integrity and reducing the risk of omissions. It also emphasizes the importance of verifying threat intelligence before applying it in practical or operational contexts.

7. Key Findings

- ATT&CK Navigator provides an effective visual interface for analyzing adversary behavior
- Layering enables structured comparison of multiple threat actors
- Version compatibility impacts feature availability (For instance, layer combination)
- Scoring and color-coding enhance interpretability of complex datasets

8. Conclusion

This exercise demonstrates practical proficiency in using the ATT&CK Navigator to analyze and compare adversary techniques. By creating, importing, and combining layers, I was able to simulate real-world threat intelligence workflows.

The ability to map and compare threat actor behaviors using the ATT&CK framework is a critical skill in modern cybersecurity operations, particularly in areas such as threat hunting, incident response, and security monitoring.